Leetcode 38 Count and Say
Brute Force

```python
class Solution:
    def countAndSay(self, n: int) -> str:
        res="1"
        for _ in range(1,n):
            new_str=""
            count=1
            for i in range(1, len(res)):
                if res[i]== res[i-1]:
                    count +=1
                else:
                    new_str+= str(count) + res[i-1]
                    count=1
            new_str += str(count)+ res[-1]
            res= new_str
        return res
```

Brute force Example Dry Run $n=4$

res = '1'
Iteration 1

res = "1"
Loop does not run range(1,1)
Final Step

new_str += "1" + "1"
new_str = "11"

res = "11"

Iteration 2

res = "11"

count = 1

Loop

i	res[i]	res[i-1]	action
1	1	1	count = 2

res = "21"

Iteration 3

res = "21"

count = 1

Loop

i	res[i]	res[i-1]	action
1	1	2	append "12"

new-str = "12"

Final Append

new-str += "11"

new-str = "1211"

return 1211

Complexity

n = length of generated string

Time Complexity

Outer Loop

n times

Inner Loop:

O(m)

String Concatenation

new_str += something

takes O(length of string)

worst case → O(n*m2)

Space Complexity → O(m)

Optimal Solution

```python
class Solution:
    def countAndSay(self, n: int) -> str:
        answer="1"
        for _ in range(1,n):
            nextAnswer=[]
            cur= answer[0]
            count=0
            for c in answer:
                if c==cur:
                    count+=1
                else:
                    nextAnswer.append(str(count) + cur)
                    count=1
                    cur=c
            nextAnswer.append(str(count) + cur)
            answer="".join(nextAnswer)
        return answer
```

Dry Run (n=4)

Start

answer = "1"

Iteration 1

cur = "1"

count = 0

Loop:

c	cur	Count
1	1	1

End, append = "1 1"

answer = "11"

Iteration 2

answer = "11"

curr = "1"

count = 0

Loop

c	action
1	count = 1
1	count = 2

Append

"21"

answer = "21"

Iteration 3

answer = "21"

curr = "2"

count = 0

Loop

c	action
2	count = 1
1	append "12", reset

Append final

"11"

answer = "1211"

Complexity

Time Complexity

Outer Loop → n

Inner Loop $O(m)$

Appending to List
$O(1)$

Join Operation $O(m)$

Total $O(n * m)$

Space $O(m)$